

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2022/2023

SEPTEMBER EXAMINATION

UCCD1024 DATA STRUCTURE AND ALGORITHMIC PROBLEM SOLVING

TUESDAY, 27 SEPTEMBER 2022

TIME : 2.00 PM – 4.00 PM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMPUTER ENGINEERING

This question paper consists of **THREE (3)** questions in Section A and **TWO (2)** questions in Section B.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**. Each question carries 25 marks.

Should a candidate answer more than **ONE (1)** question in Section B, marks will only be awarded for the **FIRST** questions in the order the candidate submits the answers.

Candidates are allowed to use a non-scientific calculator.

Answer questions only in the answer booklet provided.

Q1. (a) Based on the definition in the program below, what is the output of the following cout statements? If the cout statement causes error, then provide the reason for the error.

- (i) `cout << (*ptr).breed[5] << "\n";` (2 marks)
- (ii) `cout << ptr->colour << "\n";` (2 marks)
- (iii) `cout << ptr->next->type[1] << "\n";` (2 marks)
- (iv) `cout << cat.next.gender << "\n";` (2 marks)

```

struct Node {
    char item;
    Node *next;
    Node(char newItem){
        item = newItem;
        next = NULL;
    }
};

```

```

int main(){
    Node item1('B');
    Node item2('D');
    Node item3('A');
    Node item4('E');
    Node item5('C');
    Node *head;

    head = &item1;
    item1.next = &item2;
    item3.next = head;
    head = &item3;
    return 0;
}

```

UCCD1024 DATA STRUCTURE AND ALGORITHMIC PROBLEM SOLVING**Q1. (b) (Continued)**

- (i) Draw the linked list that is created after the segment of codes in main(). (4 marks)
- (ii) Write statement to insert node with value 'C' as the third node in the linked list created in main(). (4 marks)
- (iii) Write statement to insert node with value 'E' as the last node in the linked list after you inserted 'C' in Q1(b)(ii). (2 marks)
- (iv) Write statement to remove node with value 'D' in the linked list after you have inserted 'E' in Q1(b)(iii). (2 marks)
- (v) Write a function **void print(Node *head)** to print the content in a linked list. The head pointer is pointing to the first node in the list when pass to function. (5 marks)

[Total : 25 marks]

Q2. (a) The following questions are based on the list shown in Figure 1.0 below.

**Figure 1.0: List With Four Items**

- (i) Assume that the list in Figure 1.0 as a stack. Redraw the stack after the following statements are executed. (5 marks)


```

stack.pop(item);
stack.push(10);
stack.push(3);
if (item % 2 == 0)
    stack.push(item / 2);
else
    stack.pop(item);
stack.push(item * 2);
      
```
- (ii) Assume that the list in Figure 1.0 as a queue where the head pointer is pointing to first node with value 7 and tail pointer is pointing to last node with value 9. Redraw the queue after the following statements are executed. Show the head and tail pointers in your queue.

UCCD1024 DATA STRUCTURE AND ALGORITHMIC PROBLEM SOLVINGQ2. (a) (ii) (Continued)

```

//both item1 and item2 are type int
queue.enqueue(3);
queue.enqueue(16);
queue.dequeue(item1);
queue.dequeue(item2);
if (queue.tail->item > 10)
    queue.enqueue(item1);
else
    queue.enqueue(item2);

```

(7 marks)

- (b) Given a queue of integers and an integer of k, write a function **bool reverseFirstkElementInQ(Queue *q, int k)** to reverse the order of the first k elements of the queue, leaving the other elements in the same relative order. For example, if k = 5 and queue have the elements [10, 20, 30, 40, 50, 60, 70, 80, 90]; the output should be [50, 40, 30, 20, 10, 60, 70, 80, 90]. **You must use a stack to do the reverse of the first k elements in the queue.** Function returns false if q is empty and true otherwise. (9 marks)

- (c) Provide one real application of using Stack and Queue respectively. You need to provide justification for your example. (4 marks)

[Total : 25 marks]

- Q3. (a) (i) Sort the following integers in ascending order by using the **Mergesort** algorithm. You need to list out each step in the sorting process.

33 7 13 17 83 72 28 23 37 67 47 11

(8 marks)

- (ii) Assume there are two sorted arrays of size m and n respectively. You need to merge these two arrays to become one sorted array with size (m + n). Give one method on how to merge the two arrays to become one sorted array. Briefly explain the best-case scenario that can occur when you use this method to merge the two arrays. (4 marks)

- (b) Use Binary Search to search for the name **Tom** in the array below. Show the index values of start, mid and end for every step.

Alice	Chris	Dan	Harry	Ken	Olive	Sally	Tom
-------	-------	-----	-------	-----	-------	-------	-----

(4 marks)

UCCD1024 DATA STRUCTURE AND ALGORITHMIC PROBLEM SOLVING**Q3. (Continued)**

- (c) There are two steps involved to derive the Big-O notation from the $f(n)$. Redraw the table below by filling in the two steps involve and the final derived Big-O for each of the $f(n)$.

$f(n)$	Step 1	Step 2	Derived Big-O
$8n^5 + 9n^2$			
$6n * (\log_2 n) + \log_2 n$			
$10 + c$ (c is a constant)			

(9 marks)

[Total : 25 marks]

SECTION B

Q4. (a) The following questions are based on Binary Search Tree (BST).

- (i) Draw the BST after the following integer values are inserted to an empty BST.

8, 6, 13, 2, 17, 3, 12, 18, 15

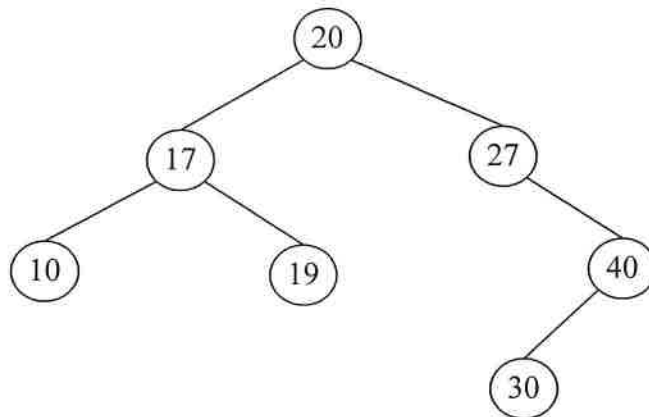
(5 marks)

- (ii) Redraw the tree after node 13 is deleted and specify which is the inorder successor node if it is case 3. (4 marks)

- (iii) The BST drawn in Q4.(a)(i) is not an AVL tree (before delete node 13). Specify which node is not balanced and specify what case is it (eg. LL, LR, LB, RR, RB, RL). Redraw the tree after the balancing. (5 marks)

- (b) Write a recursive function **void BST::largerThan(BTNode *ptr, int value)** to print all the node values in a Binary Search Tree (BST) that is larger than the variable value. Assume that ptr is pointing to root when the function is called. If variable value = 18 and the ptr is pointing to node 20 in the BST in Figure 2.0 below when function is called, it will print **19 20 27 30 40** as the output.

Q4. (b) (Continued)

**Figure 2.0 Binary Search Tree (BST)**

(7 marks)

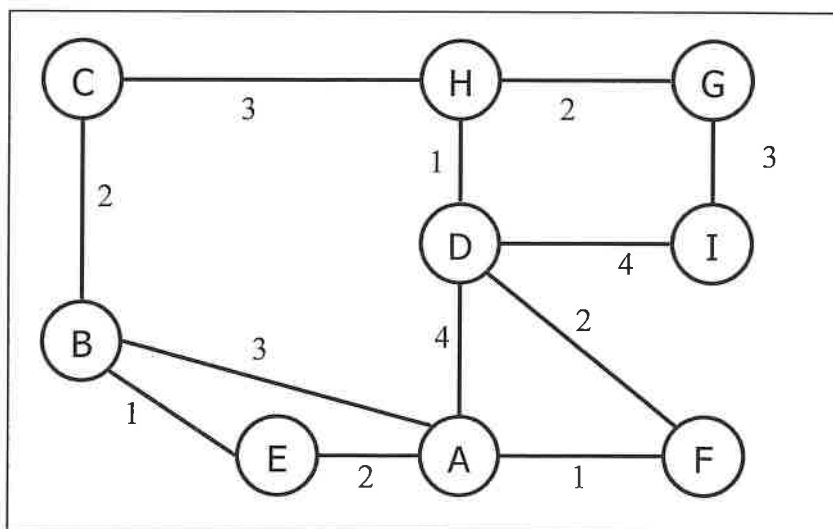
(c) Suggest the best data structures (e.g.: linked list, stacks, queues, binary tree, ...) to represent the following scenario. You need to provide justification for your choice.

(i) You want to implement a back functionality in internet browser so that you can go to the previous page. (2 marks)

(ii) You want to represent a maze in a game. (2 marks)

[Total : 25 marks]

Q5. (a) Figure 3.0 shows an undirected weighted graph A.

**Figure 3.0 Undirected Weighted Graph A**

(i) Construct an adjacency matrix representation table from the Undirected Weighted Graph A in Figure 3.0. (4 marks)

UCCD1024 DATA STRUCTURE AND ALGORITHMIC PROBLEM SOLVING**Q5. (a) (Continued)**

- (ii) Starting at vertex C, traverse the graph by using **Depth-First-Search (DFS)**. The vertices are to be traversed alphabetically. Give the order in which the vertices are traversed. (4 marks)
 - (iii) Starting at vertex C, traverse the graph by using **Breadth-First-Search (BFS)**. The vertices are to be traversed alphabetically. Give the order in which the vertices are traversed. (4 marks)
 - (iv) Draw the **Minimum Spanning Tree** for the Undirected Graph A in Figure 3.0 using Prim's Algorithm. Starting vertex is C. (5 marks)
- (b) In order to check if the elements in an array are sorted in ascending or not, we can just compare the neighboring elements in the array. If we find any of the i^{th} element that is not smaller than the $i^{\text{th}} + 1$ element, then the array is not sorted in ascending order. Implement the following functions to check if the array elements are sorted in ascending order.
- (i) Write a function **bool iterativeCheckSortedArr (int arr[], int n)** to check if all array arr elements are sorted in ascending order using **iterative (loop)** method. Variable n is the size of the array. Return true if the array is sorted and false otherwise. (4 marks)
 - (ii) Write a function **bool recursiveCheckSortedArr (int arr[], int n)** to check if all array arr elements are sorted in ascending order using **recursive** method. Variable n is the size of the array. Return true if the array is sorted and false otherwise. (4 marks)
- [Total : 25 marks]
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